



ZEESHAN ASIF

Passport: JY1880271 **Nationality:** Pakistani **Date of birth:** 12 Feb 2001

Place of birth: Lahore, Pakistan **Phone number:** (+92) 3060970771

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ABOUT ME

Research-oriented materials engineer with experience in alloy development, corrosion behavior, and electrochemical characterization. Strong background in materials processing and structure-property relationships. Interested in ceramic materials, sintering processes, and advanced manufacturing techniques. Motivated to contribute to the development of lightweight, mechanically robust, and high-performance materials.

WORK EXPERIENCE

CCRC-IMME, University of the Punjab – Lahore, Pakistan

Address: Canal Bank Rd, Punjab University New Campus, 54000 Lahore (Pakistan)

Research assistant

[4 Aug 2023 – 4 Mar 2025]

- Conducted experimental research on the corrosion behavior of magnesium alloys, with emphasis on materials processing, surface treatments, and electrochemical characterization. Applied structure-property analysis and supported laboratory training and research teams toward publication-oriented outcomes.

Research engineer

[10 Nov 2024 – Current]

- Conducted experimental research on the corrosion behavior of magnesium alloys, emphasizing materials processing, surface treatments, and electrochemical characterization, with a focus on structure-property relationships and support of laboratory research activities leading to publication-oriented outcomes.

Hydroliz (SMC Private) Limited – Lahore, Pakistan

Address: H. 06, Aftab Park, St. 03 Bund Road, Kot Kamboh Lahore, 54000, Lahore (Pakistan)

Process engineer

[2 May 2025 – Current]

- Conducting laboratory experiments on zinc recovery from zinc flue dust using hydrometallurgical techniques. Designing and assembling custom experimental setups, and performing wet-chemistry procedures. Managing procurement of materials and equipment, and maintaining detailed experimental records and reports.

Heavy Mechanical Complex – Taxila, Pakistan

Address: Hattar Road, HMC Colony, 47050 Taxila (Pakistan)

Internee Engineer

[10 Jul 2023 – 7 Aug 2023]

- Throughout my 4-week internship, I actively engaged in the responsibilities, spanning production planning and control, mechanical tasks, foundry and forge works, quality assurance and control, and technology-related duties

EDUCATION AND TRAINING

B.Sc. (Engg.) Metallurgy and Materials Engineering

University of the Punjab [7 Oct 2020 – 1 Oct 2024]

City: Lahore | Country: Pakistan | Field(s) of study: Engineering, manufacturing and construction | Final grade: 3.58 | Level in EQF: EQF level 6

Intermediate in Pre-engineering

Punjab College of Sciences [12 Mar 2018 – 8 Jun 2020]

City: Lahore | Country: Pakistan | Field(s) of study: Pre-Engineering | Final grade: A+ | Level in EQF: EQF level 4

Matriculation

Lahore Grammar School [12 Mar 2016 – 11 Apr 2018]

City: Lahore | Country: Pakistan | Field(s) of study: Sciences | Final grade: A+ | Level in EQF: EQF level 2

PROJECTS

Academic Projects

1. Prepare mullite rods (minimum dimensions 1 cm in diameter and 2 cm in height) in the lab. Demonstrate that mullite strength depends on the sintering temperature.
2. Prepare a frit glaze in the lab. Apply this glaze on a ceramic tile piece. The glaze should have blue, red or green colour.
3. To determine and study the Effect of different % of Bentonite Clay and Moisture Content on the Dry Compression Strength of moulding sand.
4. To create an assembly of components that can transform rotational motion to linear motion using a CAD software's of your choice.
5. Two or more pieces of hollow cylinder need to be manufactured using sand molding technique and aluminum alloy.

LANGUAGE SKILLS

Mother tongue(s): Urdu | Panjabi; Punjabi

Other language(s):

English

LISTENING C1 **READING** B2 **WRITING** B2

SPOKEN PRODUCTION C1

SPOKEN INTERACTION B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Microsoft Office / Canvas (learning management systems) / Echem analyst Gamry / use CAD software / MATLAB / Basic knowledge of softwares for image analysis (ImageJ, Gwyddion)

PROFESSIONAL DEVELOPMENT & WORKSHOPS

[7 Jan 2026 – 15 Jan 2026]

Future Mineral Pioneers (FMP) 2026 - Competitive International Program, Selected Participant (70+ teams from 375+), Riyadh, Saudi Arabia, 2026

Represented our project "Mine as a Battery" focusing on sustainable mineral innovation and energy storage, collaborating with international mentors in a fully funded program sponsored by the Saudi Government.

[21 Oct 2024 – 22 Oct 2024]

Attended "2 days workshop on Nanotech Essentials" organized by Department of Plant pathology, University of Punjab, Lahore 54000, Pakistan.

Gained working knowledge of software such as Echem Analyst, Origin, ChemDraw, ImageJ, and X'Pert HighScore Plus for data analysis, electrochemical interpretation, molecular drawing, imaging, and XRD pattern analysis.

[23 Sep 2024 – 25 Sep 2024]

Attended "2-day Nanotechnology Training Workshop & One day Symposium on Nanotechnology research"

Gained hands-on experience with advanced material characterization techniques, AFM, SEM, TEM, FTIR, and AAS. Participated in expert-led sessions and practical demonstrations focused on applications in nanomaterials.

PUBLICATIONS

[2025]

Investigating the role of Al₂Ca particles in the corrosion behavior of AZX311 Mg alloy in chloride and sulfate electrolytes

Authors: Hafiz Muhammad Rehan Tariq, Umer Masood Chaudry, Zeeshan Asif, Saira Niaz, Ameerq Farooq, HongBiao Dong, Tea-Sung Jun | **Journal Name:** Electrochimica Acta

H.M.R. Tariq, U.M. Chaudry, Z. Asif, S. Niaz, A. Farooq, H.B. Dong, T.S. Jun. Investigating the role of Al₂Ca particles in the corrosion behavior of AZX311 Mg alloy. Electrochimica Acta, Vol. 543 (2025), 147546.

[2025]

Influence of CaO alloying on the corrosion resistance and biocompatibility of AZ61 Mg Alloy in physiological environments

Authors: Umer Masood Chaudry, Hafiz Muhammad Rehan Tariq, Zeeshan Asif, Ameerq Farooq, Asima Tayyeb, Hongbiao Dong, Mansoor Ali, Tea-Sung Jun | **Journal Name:** Journal of Alloys and Compounds

Chaudry U.M., Tariq H.M.R., Asif Z., Farooq A., Tayyeb A., Dong H., Ali M., Jun T.S. (2025). Influence of CaO alloying on corrosion resistance and biocompatibility of AZ61 Mg alloy. J. Alloys Compd., 1048, Article 185197.

CONFERENCES AND SEMINARS

[27 Oct 2025 – 31 Oct 2025] National Centre for Physics (NCP), Islamabad, Pakistan
19th International Symposium on Advanced Materials (ISAM-2025) Participated in ISAM-2025, organized by PAMF in collaboration with IST, presenting research on "Ca-P conversion coatings on AZ31B magnesium alloy" and receiving the Best Oral Presentation Award among national and international participants.

[3 Dec 2025 – 4 Dec 2025] University of Education, Lahore, Pakistan
6th Annual International Conference on Advances in Materials Sciences (AIMS-2025) Presented an oral presentation at the 6th Annual International Conference on Advances in Materials Sciences (AIMS-2025) on "Enhancing Structural Integrity of 6061 Al alloy through Sustainable and Eco-Efficient Friction Stir Welding Practices," highlighting improvements in mechanical performance using eco-friendly processing methods.

HONOURS AND AWARDS

[31 Oct 2025] Pakistan Advanced Materials Forum (PAMF), Islamabad, Pakistan
Best Oral Presentation Award Received the Best Oral Presentation Award at ISAM-2025, NCP Islamabad, for research on magnesium alloys and corrosion behavior, recognized for the quality and impact of the presentation among national and international participants.

[3 Feb 2024] Government of Pakistan
Received Merit-Based Lenovo Laptop Award from Prime Minister of Pakistan

RESEARCH MANUSCRIPTS

Tool rotation speed driven microstructure evolution and enhancement of mechanical and electrochemical properties in friction stir welded 6051 Al-Si alloy. (Manuscript submitted to Welding in the World)

Muhammad Faizan, Zeeshan Asif, Muhammad Arslan Hafeez, Hammad Ahmed, Ibrahim Askari, Muhammad Afzal, Ameerq Farooq

RECOMMENDATIONS

Name: **Dr. Ameerq Farooq** | Research Supervisor
Mr. Zeeshan Asif has worked under my supervision as a Research Assistant at the Corrosion Control Research Cell, University of the Punjab. He is a hardworking, research-driven, and technically sound young engineer with a keen interest in innovation. I strongly recommend him for graduate-level studies.
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